



Mahavir Education Trust's

Shah & Anchor Kutchhi Engineering College

An Autonomous Institute Affiliated to University of Mumbai

Program Structure for Final Year B.Tech. in Artificial Intelligence & Data Science (with effect from 2025–2026)

Semester VII

				Learning Scheme					Credits	Assessment Scheme												
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week			SL	Total		Exam	Theory					Practical				SLA		Total
											CIA		ESE	Total		CIAP	ESEP	Total				
				CL	TL	LL					Duration in Hrs	MSE								CCE	Max	
1	Deep Learning	-	ADCR1701	3	1	2	2	120	4	2+2	20	20	60	100	40	20	30	50	20	25	10	175
2	Big Data Analytics	-	ADCR1702	3	1	2	2	120	4	2+2	20	20	60	100	40	20	30	50	20	25	10	175
3	Department Level Optional Course-3	-	ADDLOCRI703x	3	1	2	2	120	4	2	20	20	60	100	40	50	-	50	20	25	10	175
4	Department Level Optional Course-4	-	ADDLOCRI704x	3	1	2	2	120	4	-	-	50	-	50	20	50	-	50	20	25	10	125
5	Institute Level Optional Course -1	-	XXILOCR1705	3	1	-	2	90	3	3	20	20	60	100	40	-	-	-	-	25	10	125
6	Major Project -1	-	ADLR1706	-	-	10	-	150	5	2	-	-	-	-	-	40	60	100	40	-	-	100
Total				15	5	18	10	720	24		80	130	240	450		180	120	300		125		875

Abbreviations:

CL: Classroom Learning TL: Tutorial Learning LL: Laboratory Learning SL: Self Learning (Activity/Assignment/Micro-Project) CIA: Continuous Internal Assessment
CCE: Continuous Comprehensive Evaluation MSE: Mid Semester Examination ESE: End Semester Examination CIAP: Continuous Internal Assessment Practical ESEP: End Semester Examination Practical
SLA: Self Learning Assessment

Note:

1. Notional Learning hours for the semester are (CL+LL+TL+SL) hrs $\times$ 15 Weeks. 2. 1 credit is equivalent to 30 Notional hrs.
3. Self learning hours shall not be reflected in the Time Table. 4. SLA will have separate head of passing



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Department Optional Course - III

Sr. No.	Course Code	Course Name
1	ADDLOCR17031	Natural Language Processing
2	ADDLOCR17032	AI for Healthcare
3	ADDLOCR17033	Neural Network & Fuzzy System

Department Optional Course - IV

Sr. No	Course Code	Course Name
1	ADDLOCR17041	User Experience Design with VR
2	ADDLOCR17042	Blockchain Technologies
3	ADDLOCR17043	Game Theory for Data Science

Institute Level Optional Course - I

Sr. No	Course Code	Course Name
1	ECILOCR1705	Management Information System
2	CSILOCR1705	Research Methodology
3	ETILOCR1705	Disaster Management and Mitigation Measures
4	ADILOCR1705	Entrepreneurship Development and Management



Program	Final Year B. Tech Artificial Intelligence and Data Science (Semester VII)	CL	TL	LL	SL	C
Course Name: Game Theory for Data Science	Course Code: ADDLOCR17043	3	1	2	2	4
Course Type:	-					
Pre-requisite:	Probability and Statistics, Discrete Mathematics, Artificial Intelligence Fundamentals					

I RATIONALE

Game Theory is a fundamental tool in Data Science to model and analyze interactions among rational agents. The understanding of strategic behavior is essential for designing algorithms in fields such as machine learning, economics, artificial intelligence, and operations research.

II COMPETENCY

The students will be able to model strategic scenarios, analyze them using game-theoretic tools, and apply these concepts to real-world data science applications.

III COURSE OUTCOMES (COs)

After the completion of course based learning Learners will be able to,

Course Outcome Number	Course Outcome Statement
1	Analyze cooperative and non-cooperative game models in real-world strategic scenarios using simulation tools.
2	Simulate extensive form games with perfect information to develop strategy trees in agent-based environments.
3	Develop Bayesian or sequential strategies for imperfect information games using game-theoretic modeling platforms.
4	Implement a mechanism design system for social preference aggregation with transparency in computational scenarios.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme				Assessment Scheme														
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week			SL	Total	Credits	Exam	Theory				Practical				SLA		Total	
				CL	TL	LL					CIA	ESE	Total		CIAP	ESEP	Total					
							Max	CCE		Max			Max	Min			Max	Max	Max	Min		Max
1	Game Theory for Data Science	-	ADDLOCR17043	3	1	2	2	120	4	-	-	50	-	50	20	50	-	50	20	25	10	125

**V LABORATORY LEARNING OUTCOME AND SUGGESTED LIST OF EXPERIMENTS**
(Minimum 8)

Sr. No.	Laboratory Learning Outcome	Laboratory Experiment Titles	Relevant COs
1	Understand basic strategic interaction through implementation of a simple Prisoner's Dilemma.	Implement Simple Prisoner's Dilemma problem.	CO1
2	Analyze games with pure strategy Nash equilibria.	Implement game with Pure Strategy Nash Equilibrium.	CO2
3	Interpret extensive form games using tree representations.	Implement game with Extensive Form – Graphs and Trees, Game Trees.	CO2
4	Evaluate strategy dominance using strategic form representation.	Implement Strategic Form game to show elimination of dominant strategy.	CO2
5	Apply minimax theorem to determine optimal strategies in zero-sum games.	Implement game using Minimax theorem, minimax strategies.	CO2
6	Analyze decision-making in games with perfect information.	Implement game with Perfect information games.	CO3
7	Model and implement games under incomplete information.	Implement game with Imperfect-information games.	CO3
8	Understand long-term strategy via repeated games.	Implement game to show concept of Repeated Games.	CO3
9	Apply Nash equilibrium concepts in Bayesian games.	Implement Bayesian game using Nash equilibrium.	CO4
10	Integrate multiple game theory concepts in a group-based simulation.	Implementation of any complete game (e.g., Tic Tac Toe, 8-Queen) using combination of concepts.	CO4

VI THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT



Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1	1. Apply fundamentals of game theory and rational choice. 2. Analyze mathematical formulation of games and rational behavior. 3. Examine non-cooperative vs cooperative games. 4. Apply understanding of computational challenges. 5. Analyze equilibria and learning techniques. 6. Evaluate self-interest models in agents. 7. Analyze applications of non-cooperative game theory. 8. Apply mixed and pure strategies. 9. Apply solution concepts to strategic interaction. 10. Apply game theory to real-world rational decision-making. 11. Evaluate models of choice under uncertainty. 12. Apply fundamentals to strategic modeling. 13. Apply reasoning under strategic interdependence.	Introduction: cooperative and non-cooperative games 1.1 Basics of Games, Strategy, Preferences, Payoffs 1.2 Mathematical basics, Rational Choice, Game Theory 1.3 Basic solution concepts, Cooperative vs Non-cooperative games 1.4 Basic computational issues 1.5 Finding equilibria and learning in games 1.6 Self-interested agents 1.7 Application areas in AI and Data Science 1.8 Pure and Mixed Strategies 1.9 Preferences and Strategic Interaction 1.10 Rational agents and strategy formulation 1.11 Choice theory and model applications 1.12 Foundations for modeling decisions 1.13 Interactive decision-making framework	13

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Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
2	CO2	1. Apply strategic concepts to classical games. 2. Analyze Nash equilibria in strategic games. 3. Evaluate strategic firm behavior in economics. 4. Analyze price competition using game theory. 5. Apply equilibrium concepts in auction design. 6. Analyze equilibria under uncertainty. 7. Apply concepts to zero-sum games. 8. Evaluate repeated strategic interactions. 9. Analyze equilibria in extensive form games. 10. Evaluate equilibrium refinement techniques. 11. Understand computation aspects of game solutions.	Introduction: cooperative and non-cooperative games 2.1 Prisoner's dilemma, Matching pennies 2.2 Nash equilibria (Theory and illustrations) 2.3 Cournot's model of oligopoly 2.4 Bertrand's model of oligopoly 2.5 Auctions and Bidding Strategies 2.6 Mixed strategy equilibrium 2.7 Zero-sum games 2.8 Repeated games (e.g. Prisoner's dilemma) 2.9 Extensive Games with Perfect Information 2.10 Subgame perfect Nash equilibrium 2.11 Computational issues in solving games	11

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Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
3	CO3	1. Apply game theory to situations with hidden information. 2. Analyze strategies in Bayesian games. 3. Apply equilibrium concepts under uncertainty. 4. Evaluate sequential decision-making processes. 5. Model and simulate imperfect information interactions. 6. Apply repeated game analysis. 7. Analyze dilemmas in repeated settings. 8. Evaluate bargaining solutions in game theory. 9. Illustrate Bayesian game outcomes. 10. Analyze learning in games with private information. 11. Apply strategic decision models.	Introduction: Games with Imperfect Information 3.1 Games with Imperfect Information 3.2 Bayesian Games: Definitions and Motivation 3.3 Nash Equilibrium in Bayesian Games 3.4 Beliefs and Sequential Equilibrium 3.5 Strategies and Information Aspects 3.6 Repeated Games 3.7 Revisited Prisoner's Dilemma 3.8 Bargaining in games 3.9 Examples and Case Applications 3.10 Adaptive strategies in Bayesian games 3.11 Strategic learning under uncertainty	11

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Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
4	CO4	1. Apply social choice theory to group decision-making. 2. Analyze properties of aggregation mechanisms. 3. Apply mechanism design principles. 4. Design efficient outcome mechanisms. 5. Apply Vickrey and VCG mechanism to auctions. 6. Model profit-maximizing auctions. 7. Evaluate real-world applications of mechanism design. 8. Model online platform bidding. 9. Evaluate auction systems in e-commerce. 10. Apply strategic analysis to exploration problems.	Mechanism Design Aggregating Preferences 4.1 Social Choice: Formal Model, Voting 4.2 Existence of social functions, Ranking systems 4.3 Mechanism design with unrestricted preferences 4.4 Efficient Mechanisms 4.5 Vickrey & VCG mechanisms (Shortest paths) 4.6 Combinatorial auctions, Profit maximization 4.7 Applications in Computer Science 4.8 Google's sponsored search 4.9 eBay auctions 4.10 K-armed bandits	10

**VII SELF LEARNING ACTIVITIES**

Sr. No.	Activity Title	Description	Mapped CO(s)
1	Case Study on Auction Algorithms	Study and analyze real-world auction mechanisms such as Google Ads auctions, eBay listings, or VCG mechanisms. Present the advantages, trade-offs, and fairness implications.	CO3, CO4
2	Game Theory Simulation using Python	Develop a Python-based simulation of classical games like Prisoner's Dilemma, Cournot competition, or Matching Pennies. Visualize equilibrium strategies.	CO1, CO2
3	Blog Series: "Game Theory for Everyday Decisions"	Create a blog/vlog series explaining key Game Theory concepts (e.g., Nash equilibrium, strategic dominance) in the context of real-life scenarios such as traffic, sports, or voting.	CO1, CO3
4	Strategy Tracker: Multi-player Game Reflection	Engage in online multiplayer games (like chess, negotiation simulators, or strategy games). Document decisions, applied strategies, and relate them to game theory concepts.	CO2, CO4

VIII SUGGESTED COURSE ARTICULATION MATRIX

Course Outcomes (COs)	Program Outcomes (POs)											Programme Specific Outcomes (PSOs)		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	2	-	2	-	-	-	-	-	-	3	3	-
CO2	-	-	-	3	3	2	-	-	-	-	-	3	3	-
CO3	-	-	-	3	3	2	-	-	-	-	-	3	3	-
CO4	-	-	3	-	-	2	3	-	-	-	-	-	3	3

IX SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS**Text Books**

Sr. No.	Title	Edition	Authors	Publisher	Year
1	An Introduction to Game Theory	-	Martin J. Osborne	Oxford University Press	-
2	Theory of Mechanism Design	-	Debasis Mishra	Indian Statistical Institute	-

**Reference Books**

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Game Theory	–	M. Machler, E. Solan, S. Zamir	Cambridge University Press	2013
2	Algorithmic Game Theory	–	N. Nisan, T. Roughgarden, E. Tardos, V. V. Vazirani (Eds.)	Cambridge University Press	2007
3	Games of Strategy	2nd Edition	A. Dixit, S. Skeath	W W Norton & Co Inc	2004
4	Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations	–	Yoav Shoham, Kevin Leyton-Brown	Cambridge University Press	2008
5	Game Theory in Wireless and Communication Networks	–	Zhu Han, Dusit Niyato, Walid Saad, Tamer Basar, Are Hjorungnes	Cambridge University Press	2012
6	Game Theory and Mechanism Design	–	Y. Narahari	IISC Press, World Scientific	–

Online References

Sr. No	Website Name and Description
A	https://nptel.ac.in/courses/106/104/106104119/ NPTEL Course: Game Theory by Prof. Partha Pratim Chakrabarti (IIT Kharagpur)
B	https://www.coursera.org/learn/game-theory Coursera: Game Theory – offered by Stanford University and University of British Columbia
C	https://gametheory101.com/ GameTheory101: Educational videos and notes on classical and advanced game theory concepts by William Spaniel
D	https://www.edx.org/course/intro-to-game-theory edX: Introduction to Game Theory by The University of Tokyo